

PCB, lot of noise issues from other components like the battery, LCD or the touch sensors have an impact on the RF signalling. But as we will move towards modular, we will actually isolate all of the RF components from the other parts of the system, so modular will actually make the performance better."

Other challenges

The whole project of modular smartphone is a big challenge. Most mobile phone manufacturers are focused on reducing sizes of the phones for better user experience and looks. The current integrated design allows designers to innovate in areas such as curved displays or higher-resolution cameras.

But when we talk about a mobile phone, a major concern is the form factor. Making the device more modular will definitely impact on size of the device. We can easily predict that,

with a modular architecture size of the phone is only going to increase. So, keeping the form factor same for the modular device will be a challenge for the designers and engineers.

The other issues that should be taken care of while designing a modular device would be liability and overall ruggedness of the modules. What will happen if the device is dropped? If it starts coming apart then we have an issue. Additionally, the modules and interconnects must be rugged enough to handle frequent plugging and unplugging.

Besides, today's non-modular devices are made to resist mechanical shocks in terms of functionality as well. Ensuring the same thing with the modular device will be a bigger challenge. We need to engineer for the fact that mechanical shocks have least impact on the performance of the individual modules as well as the overall body.

A feasible technology standpoint

Modular architecture in smartphones is challenging but it opens up a new area of research which is probably going to be very exciting for the engineers. It is definitely a feasible technology standpoint but the associated considerations must not be sidelined.

When a device becomes mobile or mobility oriented, we take care of many issues. But if we are going modular, we are actually increasing those engineering challenges that are not going to be very easy to solve.

The key focus in the modular designs would be to ensure there are common communication and interface mechanisms at both hardware and software levels. As long as the modules adhere to these specifications, these issues would be addressed. ●

The author wrote this as a technical journalist at EYF. He recently opted and shifted to the hands-on training division of EYF

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